

mcpy: Grand Canonical Monte Carlo for atomistic systems with machine-learning potentials

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

mcpy is a Python package for grand canonical Monte Carlo (GCMC) simulation of atomistic systems of any kind (surfaces, nanoparticles, fluids, and bulk) at controlled temperature and chemical potential. It is built on the Atomic Simulation Environment (ASE) (Larsen et al., 2017), so any ASE-compatible calculator can drive the sampling: density-functional theory codes, classical force fields, or machine-learning interatomic potentials (MLIPs) such as MACE (Batatia et al., 2022). mcpy implements both GCMC and replica-exchange GCMC (Swendsen & Wang, 1986) in a single, modular run loop, and runs with or without local relaxation of trial moves. Following the hybrid scheme of Senftle et al. (2014), insertions and deletions can be relaxed locally before the acceptance test, which makes sampling efficient in densely packed metallic systems where rigid insertions would almost always be rejected; relaxation can equally be switched off for standard GCMC of fluids and other systems. The same loop runs on CPUs via MPI or on GPUs through the NVIDIA Alchemi backend, which relaxes a batch of replicas in a single evaluation. The package provides modular trial moves (insertion, deletion, displacement, permutation, shake, and Brownian; permutation enables chemical-ordering optimization in multi-component systems), configurable cell geometries (periodic box, rectangular sub-slab, spherical region around a nanoparticle, and user-defined cells), and post-processing utilities that turn raw ensembles into surface and nanoparticle phase diagrams.

Statement of need

Predicting the equilibrium composition of surfaces and nanoparticles under reactive atmospheres requires sampling over particle number (the grand canonical ensemble) at near-first-principles accuracy. MLIPs now deliver close-to-DFT accuracy at a small fraction of the cost, making large-scale GCMC of realistic structures increasingly practical. Yet no open-source package combines relaxation-coupled GCMC, replica exchange, an ASE-native MLIP backend, and the cell geometries needed for finite nanoparticles in a single workflow.

mcpy is aimed at computational chemists and materials scientists studying heterogeneous catalysis, oxidation, and gas-surface equilibria, where the stable composition, not just a fixed structure, is the quantity of interest. Although it was developed for these applications, the grand-canonical sampling is system-agnostic: it runs on any system ASE can represent and any ASE-compatible calculator, from molecular fluids to bulk solids, so the same workflow extends well beyond the metallic surfaces and nanoparticles that motivated it.

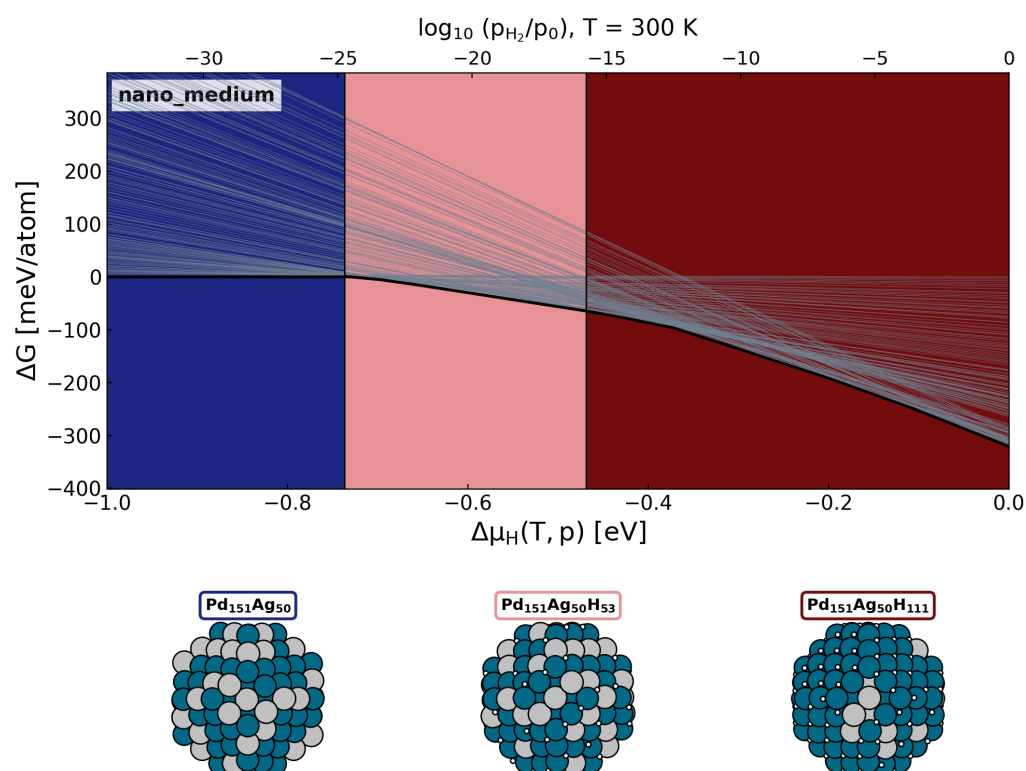


Figure 1: Hydrogenation phase diagram of a 201-atom palladium–silver nanoparticle ($\text{Pd}_{151}\text{Ag}_{50}$) obtained with replica-exchange GCMC in *mcpy*, driven by the small AGNESI MACE potential (Batatia et al., 2025). The free energy of formation per atom ΔG (in meV/atom) is shown as a function of the hydrogen chemical potential $\Delta\mu_{\text{H}}$ (equivalently $\log_{10}(p_{\text{H}_2}/p_0)$ at $T = 300$ K); the lower band shows the most stable $\text{Pd}_{151}\text{Ag}_{50}\text{H}_z$ structure sampled in each chemical-potential window, from the bare particle at low μ_{H} to a hydrogen-loaded $\text{Pd}_{151}\text{Ag}_{50}\text{H}_{111}$ at high pressure. The chemical ordering of the bare $\text{Pd}_{151}\text{Ag}_{50}$ particle was first optimized with a single basin-hopping, replica-exchange run over six temperatures; hydrogenated structures were then sampled by replica-exchange GCMC at hydrogen chemical potentials from $\Delta\mu_{\text{H}} = -1$ to -0.35 eV with six temperatures per chemical-potential window.

State of the field

Several established codes perform Monte Carlo sampling of atomistic systems, but each covers only part of the space *mcpy* targets. RASPA2 (Dubbeldam et al., 2016) and its recent C++ rewrite RASPA3 (Ran et al., 2024) are specialized for adsorption and diffusion of molecules in nanoporous materials with classical force fields. DL_MONTE (Brukhno et al., 2021) is a general-purpose Monte Carlo code that supports GCMC of solids and fluids, but with classical interaction models and its own input ecosystem rather than native integration with MLIPs or ASE. LAMMPS (Thompson et al., 2022) provides a GCMC fix, parallel tempering, and machine-learning pair styles, but it neither couples replica exchange to grand-canonical sampling nor exposes a relaxation-coupled GCMC in which an arbitrary ASE calculator drives each acceptance test. ASE itself (Larsen et al., 2017) supplies the Atoms object and a unified calculator interface spanning DFT codes, classical force fields, and MLIPs, but no grand-canonical ensemble built on top of it.

We built *mcpy* as a new package rather than extending any one of these because the capabilities it unifies live in different ecosystems: the calculator-agnostic accuracy of ASE, the grand-canonical and replica-exchange machinery of dedicated Monte Carlo codes, and the geometric flexibility needed for finite nanoparticles. Adding MLIP-native, relaxation-coupled GCMC to RASPA2 or DL_MONTE would require reimplementing ASE's calculator abstraction, while

55 wrapping LAMMPS in a relaxation-coupled, ASE-native grand-canonical loop with free-volume
 56 cell geometries and *ab initio* thermodynamics post-processing would reproduce much of mcpy's
 57 design around an MD-centric engine. Building directly on ASE instead lets mcpy inherit the
 58 entire ASE calculator ecosystem and contribute the missing grand-canonical layer. That layer
 59 is the package's distinct contribution: ASE-native, MLIP-driven, relaxation-coupled GCMC
 60 with replica exchange and nanoparticle/surface geometries.

61 Software design

62 mcpy is organized around four extensible abstractions: *ensembles* that drive the Monte Carlo
 63 loop (GrandCanonicalEnsemble, CanonicalEnsemble, ReplicaExchange), *moves* that propose
 64 trial configurations (insertion, deletion, displacement, permutation, shake, and Brownian),
 65 *cells* that define the sampling region and estimate its free volume (periodic box, rectangular
 66 sub-slab, spherical region around a nanoparticle, and user-defined cells), and *calculators* that
 67 wrap ASE energy evaluations. This separation reflects the central design choice: by expressing
 68 the physics through ASE's Atoms and calculator interfaces, mcpy trades the raw speed of a
 69 monolithic, hand-tuned Monte Carlo kernel for the ability to drive identical sampling code
 70 with any ASE-compatible backend, from a DFT code to a MACE potential (Batatia et al.,
 71 2022). Because the dominant cost in the target application is the energy evaluation rather
 72 than the Monte Carlo bookkeeping, that trade-off is strongly favorable, and it lets a researcher
 73 exchange accuracy for cost without rewriting the simulation.

74 The second design choice is an optional relaxation-coupled acceptance criterion. Following the
 75 hybrid scheme of Senftle et al. (2014), each trial insertion or deletion can be relaxed locally (a
 76 short LBFGS minimization) before the acceptance test. This adds cost to each step, but in
 77 densely packed metallic surfaces and nanoparticles a rigid insertion overlaps neighboring atoms
 78 and is almost always rejected; the local relaxation is what makes grand-canonical sampling
 79 converge at all for these systems, so the per-step cost is the enabling trade-off rather than an
 80 overhead.

81 The third choice concerns ergodicity and parallelism. Replica exchange (Swendsen & Wang,
 82 1986) across temperatures or chemical potentials improves mixing, and mcpy exposes two
 83 complementary parallelization paths: over MPI ranks on CPUs, one replica per rank, or on a
 84 GPU through the NVIDIA Alchemi backend, which relaxes a batch of replicas together in a single
 85 evaluation. The MPI path is portable and dependency-light (mpi4py is optional); the GPU path
 86 trades that portability for throughput on large systems where batched relaxation dominates.
 87 Because the ensemble loop is decoupled from the calculator, the same replica-exchange code
 88 runs under either path.

89 Finally, the modular loop is reused beyond equilibrium sampling: every ensemble doubles as a
 90 basin-hopping global optimizer, emitting a running trajectory of strictly improving (minimum-
 91 energy) configurations, and replica-exchange runs additionally report the global minimum
 92 found across replicas. Compound moves that perturb several atoms per trial make these
 93 basin-hopping escapes effective in densely packed systems. Simulations log per-step energies,
 94 particle counts, and per-move acceptance ratios, and write extended-XYZ trajectories alongside
 95 an optional minima trajectory of strictly improving structures. The bundled phase-diagram
 96 utilities map sampled ensembles onto *ab initio* thermodynamics phase diagrams (Reuter &
 97 Scheffler, 2001), connecting the simulated chemical potential to experimental temperature
 98 and partial pressure.

99 To validate the grand-canonical sampling independently of any MLIP, we reproduced two
 100 Lennard-Jones reference systems against LAMMPS (Thompson et al., 2022) (Figure 2): a
 101 purely repulsive fluid in reduced units and real argon in physical (eV) units. The average
 102 particle number and density as a function of chemical potential agree between the two codes
 103 across both unit systems, including the sharp gas-liquid condensation of argon. Although mcpy
 104 has so far been applied mainly to metal surfaces and nanoparticles, this benchmark confirms

that its grand-canonical machinery is correct and general, applicable to atomistic materials beyond the metallic systems that motivated it.

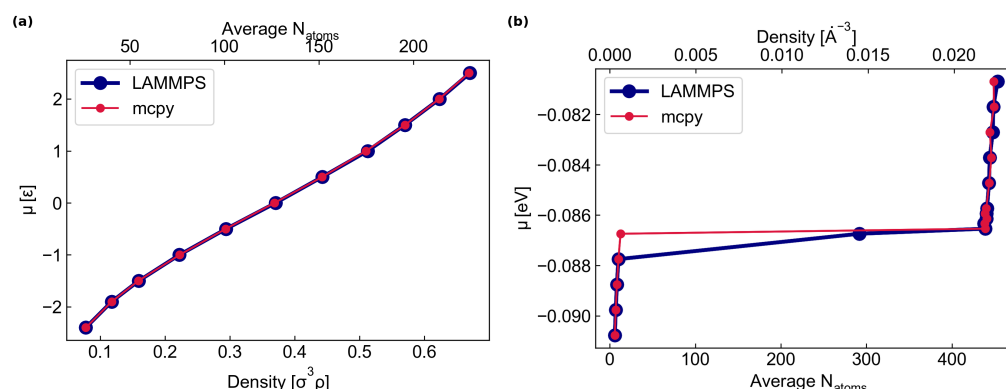


Figure 2: Validation of mcpy's grand-canonical sampling against LAMMPS (Thompson et al., 2022) for two Lennard-Jones reference systems. (a) A purely repulsive Lennard-Jones fluid in reduced units: chemical potential μ (in units of ϵ) versus reduced density $\sigma^3 \rho$ and average particle number. (b) Real argon in physical units: μ (eV) versus average particle number and number density, capturing the sharp gas-liquid condensation. mcpy (red) reproduces LAMMPS (blue) across both unit systems.

Research impact statement

mcpy was developed to study the equilibrium oxidation of metal nanoparticles and surfaces under reactive atmospheres at near-DFT accuracy. The replica-exchange GCMC workflow was built during doctoral research at the Universitat de Barcelona and underpins a forthcoming manuscript on the oxidation thermodynamics of silver nanoparticles, with further application to additional catalytic metal/oxide systems within the IQTCUB group. Figure 1 shows a representative result from this workflow: the hydrogenation phase diagram of a 201-atom Pd-Ag nanoparticle, in which mcpy recovers the progressive hydrogen loading of the particle as the chemical potential increases. By coupling MLIP-driven energetics with grand-canonical sampling and *ab initio* thermodynamics post-processing, mcpy predicts stable surface and nanoparticle compositions as a function of temperature and chemical potential, quantities that are otherwise inaccessible to fixed-composition relaxations. The package is open-source, documented with worked examples covering GCMC, replica exchange, and phase-diagram construction, and continuously tested across Python 3.11–3.13, so that the published workflows can be reproduced and extended by other groups.

Acknowledgements

The authors received no specific funding for this work.

AI usage disclosure

Claude (Opus) assisted with code refactoring, test scaffolding, packaging configuration, and editing of the manuscript and documentation. All AI-assisted outputs were reviewed, edited, and validated by the human authors, who designed the overall code architecture and take full responsibility for the accuracy and originality of the software and all submitted materials.

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